Supplement S1: review guide

Inference Delay and Deadline Handling in Thermal Controller Evaluation: A Retrospective Simulation Study

Revision of 19 October 2026. Read together with the revised manuscript and the executable Supplement_S1 ZIP. This guide provides the six supplementary figures and accessible summaries of the results moved from the main text.

Reproduce the saved-result analyses

```
python -m pip install -r requirements.txt
python -B scripts/reproduce_review.py
```

Python 3.10+, NumPy, and Matplotlib are required. The command verifies frozen file hashes, paired contrasts, method details, candidate margins, command restoration, thermal responses and Small initialization summaries. It writes five main figures and six supplementary figures to outputs/. It calls no training, inference, GPU, or plant simulator.

Main item	Evidence directory under data/	Output / script
Tables 5-6	primary/	Historical table6/table7 CSVs; scripts/reproduce.py
Table 7	revision_20261013/neartie/	Eligible-score threshold and episode summaries
Table 8 / Fig. 3	replay/	57 paired contrasts; scripts/reproduce_replay.py
Tables 9-10 / Fig. S5	revision_20261013/mechanism/	Event, episode, setting and thermal summaries
Table 11	numerical/	Historical table8 CSV; scripts/reproduce.py
Table 12 / Fig. S6	small_seed/	Small initialization checks
Tables 13-14 / Fig. 5	monte_carlo/	reproduce_monte_carlo.py

TABLE_FIGURE_MAP.md gives the complete mapping. METHODS.md documents model inputs, training/selection, candidate masks, normalization and post hoc definitions. Source fragments under methods/ retain the moved equations and detailed protocols. Historical file numbering is preserved for provenance.

Scope of reproducible evidence

This package contains saved results, compact inputs, and author audit receipts. Full weights, training inputs, raw timing journals, quadrature arrays and temperature fields are omitted. It supports reanalysis and figure regeneration, not independent new closed-loop execution. Three Small initialization repeats are now included; one fails the validation improvement criterion and is retained. The 64 Monte Carlo repetitions are included. Episode-bootstrap and timing-repetition intervals have distinct units and remain conditional on the specified models, pools and design.

Development and selected-case evidence

Cohort	Use and interpretation
A	Training/checkpoint diagnostic on eight reused validation episodes; six nominal and two initially hot.
B	Validation-based checkpoint selection in five scorer families; selection trajectories are reused.
C	A different frozen direct scorer and one post hoc selected episode; supporting numerical diagnostic.
D	Primary frozen systems: 40 train, 16 validation, eight calibration, 32 test and 16 OOD episodes. Primary test comparisons use 24 nominal episodes; eight initially hot episodes are separate.

A: offline and downstream criteria

Extending the direct scorer from 250 to 1,000 total steps reduced mean FV candidate-cost difference from 5.6138 x 10^-7 to 1.4751 x 10^-7. Mean closed-loop cost increased from 0.0299374 to 0.0480506 on the same eight validation episodes. Five of six nominal episodes worsened and two hot episodes tied. Neither comparison had deadline misses. The learning-rate schedule also changed; this is not an isolated training-duration effect.

B: selection on reused validation episodes

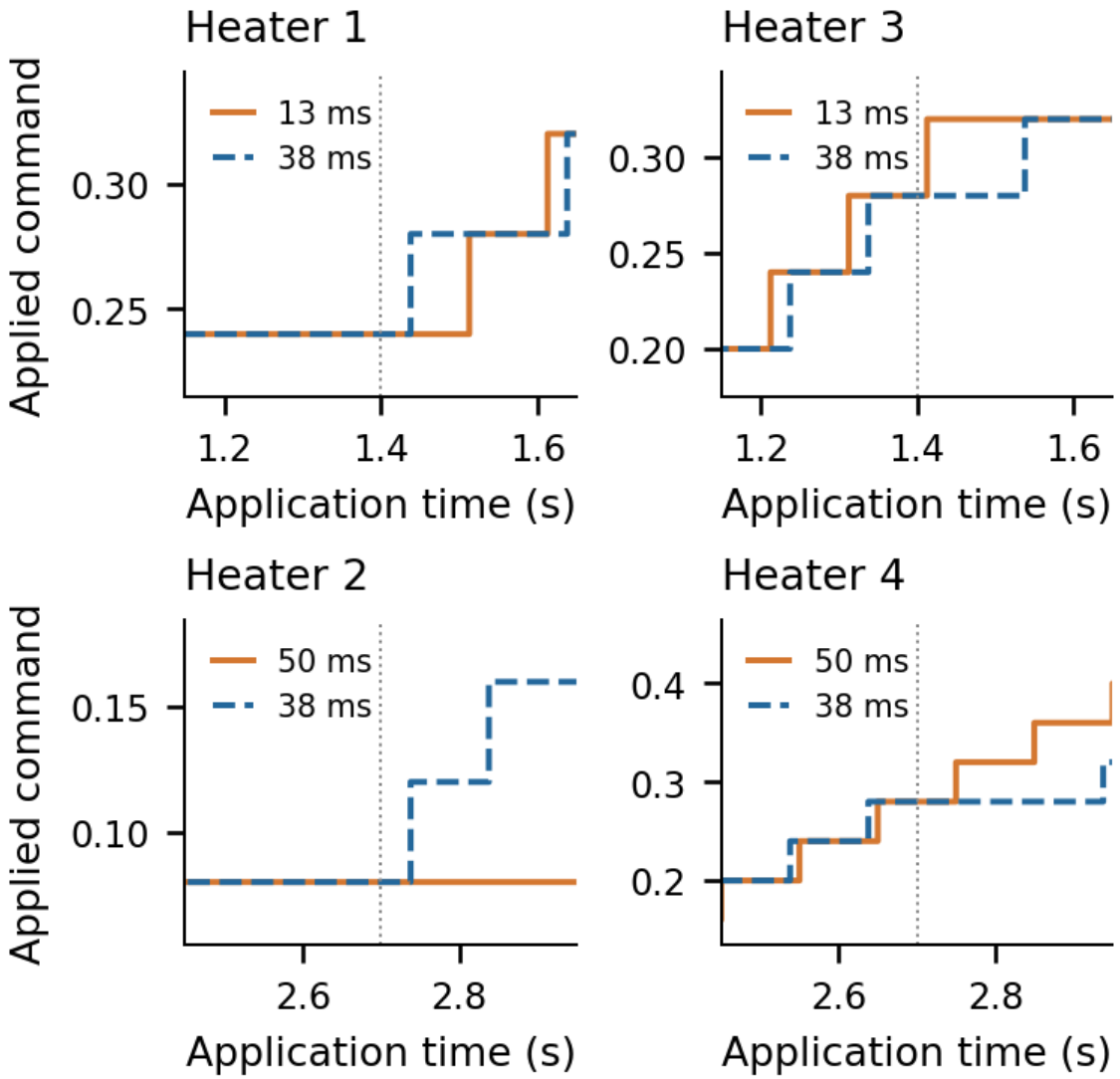
Offline and rollout criteria selected different checkpoints in all five families. The rollout-selected nominal validation costs were 5.09-68.17% lower, despite larger candidate-cost differences. A sensor-only checkpoint selected offline produced new limit violations in 11 of 40 validation trajectories spanning three episodes. Only rollout-selected policies received the separate test evaluation. These data cannot establish independent selector benefit without testing the unselected alternatives.

C: selected sensitivity diagnostic

For the selected episode at a 10-ms plant cap, 13-, 38- and 50-ms normal-path delays produced costs 0.0362894, 0.0099116 and 0.0366019 with no deadline misses. The 38-ms case was selected because it accounted for 100.71% of net degradation across nine fresh nominal confirmation episodes. This selection does not define a representative episode or an optimal delay.

The first differing requests occur at 1.4 and 2.7 s with matched eligibility masks and no gate intervention. Figure S1 shows applied commands. Figure S4 contrasts fixed-command integration with recomputed decisions. Full details are in methods/development_detail.tex, development_and_selected_C_results.tex, selected_C_detail.tex and cohorts_and_procedure.tex.

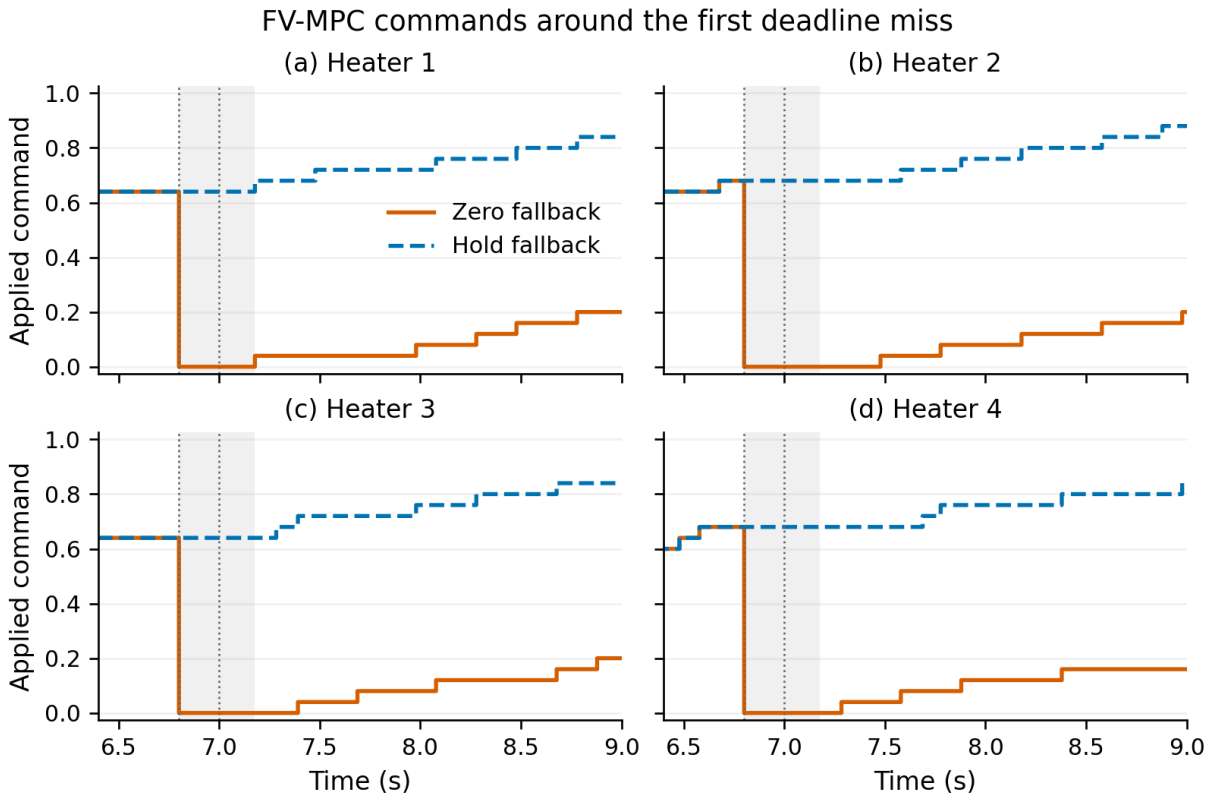
Figure S1. Selected C command diagnostic



Archived applied-command comparison for the selected C episode. Individual command vectors differ before their request-aligned means do. Source commands, first differing requests, score margins and case provenance are retained in data/selected_C/ and methods/selected_C_detail.tex.

C uses a different frozen direct scorer from D and was selected after examining outcomes. It supports a conditional implementation diagnostic; it does not estimate population performance, establish chaotic dynamics, or identify an optimal physical delay.

Figure S2. First deadline miss in D



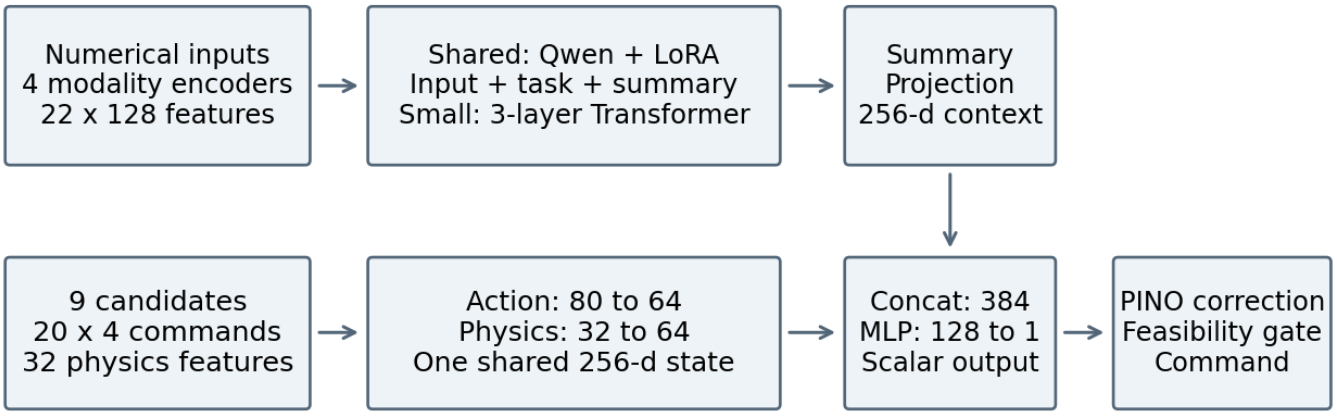
Four-heater applied commands for episode l2_240070 under original replay at a 10-ms cap. Orange: zero fallback; blue: hold. Expirations occur at 6.8 and 7.0 s; shading ends at the next nonfallback application, 7.177863943 s. Curves use actual application times and right-continuous holds. The episode is unchanged from main Figure 4, selected by lower-median zero-delay FV cost.

The pre-expiration vector [0.64, 0.68, 0.64, 0.68] resets to zero or remains unchanged under hold. The next nonfallback vectors are [0.04, 0, 0, 0] and [0.68, 0.68, 0.64, 0.68], respectively. Because one ordinary decision changes only one coordinate by .04, full-vector restoration from zero requires at least 66 increasing moves. This event is censored at the next expiration .2 s later; the historical vector need not remain thermally desirable.

Across all seven fixed settings, 715 zero-policy expirations reset nonzero vectors: 324 restore before follow-up ends and 391 are censored. Another 413 expirations occur at an already zero vector. Observed-only times are not overall recovery estimates; events within an episode are dependent.

Both illustrated cases contain 592 request and application/deadline records. The command-increment penalty retains N=600 nominal slots, regardless of skipped requests or unchanged commands. The reconstructed values, 0.00001872 (zero) and 0.00002666666666666667 (hold), match the stored metrics.

Figure S3. Original frozen residual systems



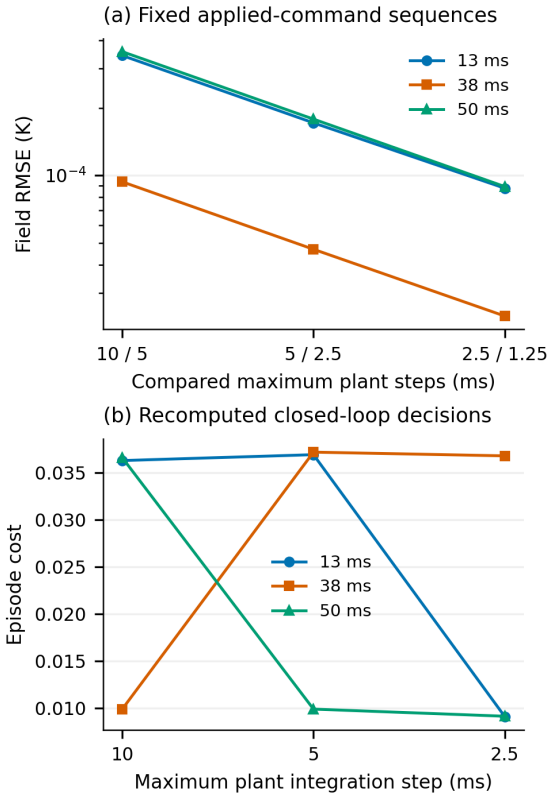
One state per request, shared by all nine candidates. No autoregressive generation.

Setting	D residual systems
Inputs / target	Same common examples and batch order; observed-state FV held-action cost over 2 s.
Optimization	AdamW, batch 16, weight decay .01, gradient clipping 1; new-module learning rate 1e-4 and LoRA adaptation rate 3e-5.
Executed / selected	50 warmup + 400 adaptation steps executed. Both select warmup step 50.
Strength selection	Alpha in {0, .5, 1}; validation at zero and 38-ms delays. Shared selects 1; Small selects .5.

At the selected Shared checkpoint the released backbone and LoRA remain frozen; newly trained thermal encoders and the replaced head supply adaptation. The full budget includes later updates absent from the tested checkpoint. Architecture, pretraining and selected alpha differ, so this comparison does not isolate transfer benefit. Model-card status and the absence of a pretraining ablation are explicit in the manuscript.

The complete residual and PINO losses, masks, target scales, retained checkpoints, layer dimensions and PI updates are provided in METHODS.md and the corresponding methods/ fragments.

Figure S4. Selected C numerical sensitivity



(a) Full-field RMSE between successive refinements, sampled at 1 Hz with the original applied commands fixed. (b) Costs with closed-loop decisions recomputed at each cap. Colors denote normal-path delays. The lowest-cost delay changes from 38 to 50 to 13 ms as the cap decreases from 10 to 5 to 2.5 ms.

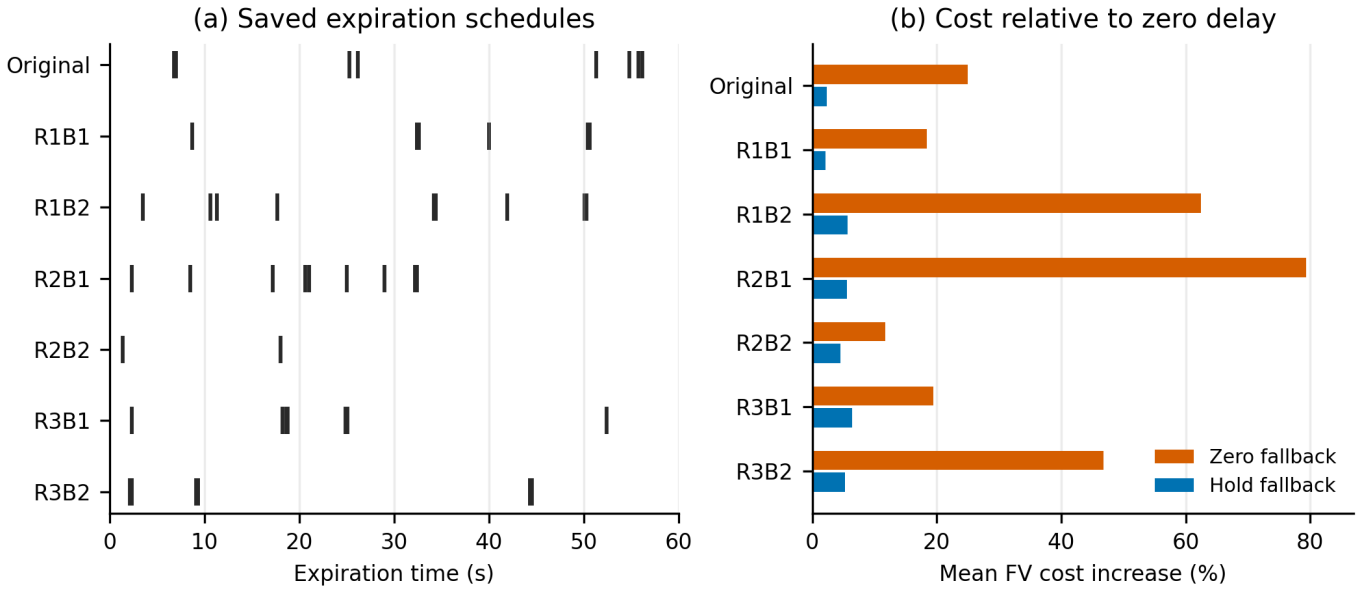
With fixed commands, cost changes from 10 to 1.25 ms remain below .0032%; the largest sampled field RMSE is 6.25 x 10^-4 K and maximum pointwise difference is 3.62 x 10^-3 K. These are empirical comparisons, not error bounds against a continuous-time solution. Small fixed-input changes coexist with larger changes when discrete decisions are recomputed.

D cost-component contrasts moved from the main text

Difference	FV replay-zero	Shared-Small zero	Shared-Small replay
RMSE (K)	1.0596	-0.0110	0.0232
Tracking (10^-5)	339.978	-2.735	5.283
Terminal (10^-5)	-8.116	-1.129	-0.220
Command (10^-5)	-4.063	0.038	0.161
Increment (10^-5)	1.981	0.002	0.001

D contrasts use 24 paired nominal episodes, original zero-fallback replay and a 10-ms cap. Full precision is retained in data/revision/physical_contrasts.csv. The penalty for abrupt increments contributes little directly; subsequent tracking changes dominate the FV total.

Figure S5. Timing placement and conditional loss



(a) Saved expiration times. Within each setting, all 24 episodes and both policies have identical request/ready/deadline schedules. (b) FV mean cost increases relative to zero delay for the same episodes under zero and hold fallback. All conditions use a 10-ms cap.

The earliest first miss occurs at 1.4 s in R2B2, which has the smallest zero-policy increase, 11.70%. R2B1 first misses at 2.3 s and has the largest increase, 79.34%. Counts, spacing, pre-expiration commands and evolving thermal states differ together. These seven settings do not identify a separate causal effect of miss placement. They also do not represent independent hardware measurements or a distribution of new timing realizations.

Recovery definitions and censoring

Full-vector restoration is the first later application at which every command component regains its pre-expiration level, within 1e-8. Follow-up stops at the next expiration or 60 s. The positive-coordinate-move lower bound ranges from 1 to 100. The sum across four coordinates is required because each candidate changes only one heater. A nominal-slot equivalent is not a wall-time bound.

The compact input package contains 336 FV application traces, 196,752 ordinary transitions and 2,256 policy-expiration records. All ordinary transitions obey the one-coordinate .04 increment rule. Reproduction uses stored commands; raw source-seal and schedule audits are retained as author receipts.

Thermal responses and score margins

Table S1. All eleven original response conditions

Method / mode	Late MAE (K)	Overshoot (K)	Settled / 24	Time* (s)
FV / replay, hold	1.089	4.641	18	37.64
FV / replay, zero	0.511	2.130	21	28.00
FV / zero delay	0.998	4.349	18	36.34
PI / replay, zero	2.774	4.801	7	50.53
PI / zero delay	2.773	4.799	7	50.56
PINO / replay, zero	1.127	4.606	18	37.67
PINO / zero delay	1.103	4.572	19	38.73
Shared / replay, zero	1.157	4.645	18	37.76
Shared / zero delay	1.080	4.512	19	38.09
Small / replay, zero	1.116	4.565	18	37.21
Small / zero delay	1.089	4.536	19	38.25

*Time is the mean among observed settlements only, conditional on the displayed settled count. Remaining episodes are censored at 60 s. Settling requires all later 10-Hz samples to remain within +/-1 K. Late MAE averages absolute spatial-mean error over 50-60 s; overshoot is the nonnegative maximum sampled mean-temperature excess. These measures do not certify asymptotic settling or local/intersample peaks.

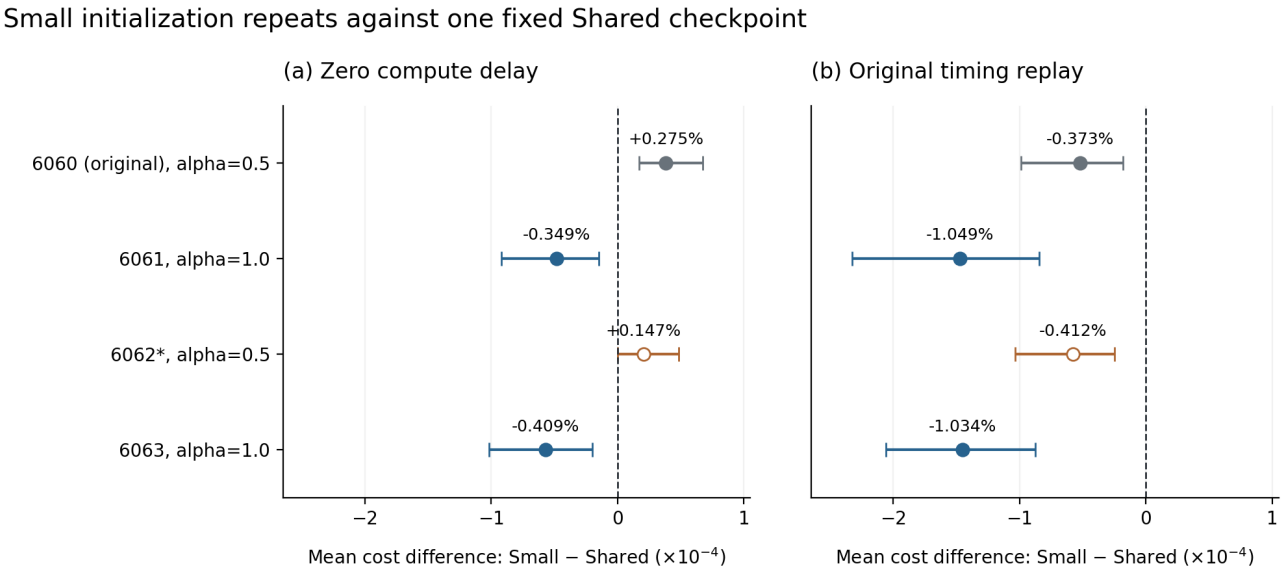
Zero fallback yields lower late error and overshoot than hold in the original replay, despite a higher whole-episode cost. The mean initial error is 26.142 K. The mean episode-wise fraction of squared mean-temperature error accumulated in 0-10 s ranges from 80.46% to 86.99% across these conditions. This is not a decomposition of the full-field cost.

Table S2. Eligible corrected-score margins

System / mode	Median	Gap <= .001	Gap <= .01
Shared / zero delay	.0735	1.70%	15.04%
Shared / replay	.0953	1.84%	15.32%
Small / zero delay	.0766	1.93%	15.45%
Small / replay	.0775	1.89%	15.51%

Each group has 14,376 scored requests and 24 initial unscored fallbacks. Eligible sets contain 5-9 candidates; no exact top-score ties occur. Scores are the dimensionless corrected values before probability-temperature scaling. Thresholds are descriptive and do not estimate solver error. These dependent requests do not establish seed robustness or a chaotic mechanism. All definitions are post hoc.

Figure S6. Small initialization sensitivity



Points: mean paired cost difference; bars: conditional 95% episode-bootstrap intervals (24 episodes, 20,000 resamples).
Percent labels use the corresponding Shared mean as denominator. Negative values indicate lower Small cost.
* Seed 6062 failed the validation regret criterion and is retained. Intervals exclude uncertainty across training seeds and timing streams.

The additional models use the original timing replay, zero fallback, a 10-ms plant cap, and the same 24 nominal episodes. Points are mean Small-minus-Shared costs; intervals resample the 24 matched episodes 20,000 times. Percent labels use Shared as denominator. These are per-checkpoint conditional intervals, not intervals over training seeds or timing distributions.

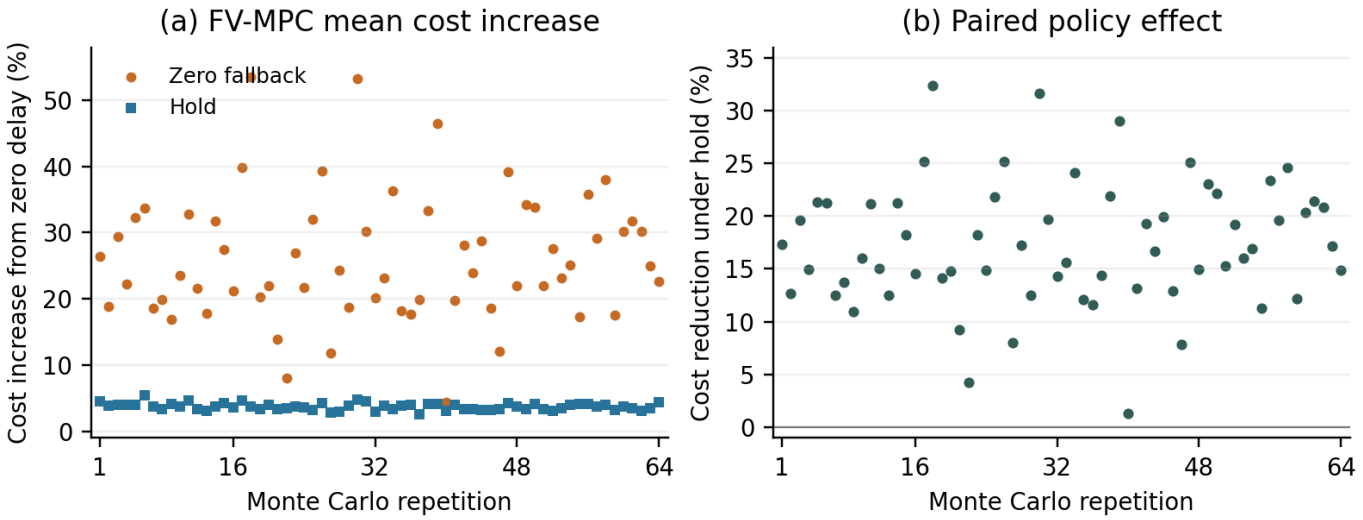
Seed	Alpha	Validation difference*	Quality	Selected step
6061	1.0	5.1677e-7	Pass	Warmup 50
6062	0.5	6.3398e-7	Fail	Warmup 50
6063	1.0	5.2181e-7	Pass	Warmup 50

*Final cached validation FV candidate-cost difference. The initial reference is 6.2609e-7. Infeasible-selection and unexpected-fallback counts are zero for all three. Alpha is selected by a distinct closed-loop validation rule. Seed 6062 is retained despite its failed improvement criterion. All 450 optimizer steps were completed for every seed.

All three additional models have lower mean replay cost than the fixed Shared checkpoint; only 6062 exhibits an ordering reversal. Seeds 6061/6063 already favor Small at zero delay. Replay shifts the cost difference toward Small in every repetition, while the zero-delay ordering depends on initialization and validation selection. The observed reversal count is not a population probability.

The compact records support recomputation of 192 test outcomes, 288 validation-selection outcomes, quality flags and paired intervals. The raw-cost/source/reload audits are author receipts. Weights, full training inputs and raw trajectories are omitted. No new timing benchmark, recalibration, Shared repetition or new-seed integration refinement was performed.

Monte Carlo timing repetitions



Main Figure 5. Each point is a 24-episode mean from one of 64 independently sampled timing repetitions. Panel (a) divides by FV zero-delay mean cost; panel (b) divides the paired zero-minus-hold difference by the zero-fallback mean in that repetition. No smoothing or confidence band is applied.

Condition	Mean increase (%)	2.5th percentile	97.5th percentile
PI / zero	.320	.090	2.757
FV / zero	26.012	10.185	49.400
PINO / zero	.436	.223	3.141
Shared / zero	.878	.739	.911
Small / zero	.250	.230	.281
FV / hold	3.722	2.842	4.712

Percentiles use linear interpolation between ordered values and describe the empirical distribution of repetition-level cost increases, not confidence intervals for a population mean. Each method uses its own zero-delay baseline. All six conditions reuse the same 24 episodes at a 10-ms plant cap.

The original Small checkpoint is used here. The three additional Small initializations use original replay only; they are not a crossed seed-by-timing experiment. IID sampling removes serial timing correlation and cannot establish deployment robustness.

Monte Carlo ranking and audit scope

Event (24-episode mean)	Count / 64	Wilson 95% interval
FV highest cost with zero	61 / 64	87.10-98.39%
FV rank 1 with hold	64 / 64	94.34-100%
FV hold lower than zero	64 / 64	94.34-100%
Original Small lower than Shared	64 / 64	94.34-100%

Other methods retain zero fallback when FV uses hold. Wilson intervals estimate conditional repetition-event probabilities under the specified IID empirical timing model. They do not cover hardware variation, unseen episodes, retraining, or finite-pool uncertainty. The three exceptions to FV being highest-cost under zero are retained.

Within the 1,536 episode-level FV pairs, hold wins 1,034, ties 114 and loses 388, with an absolute cost tolerance of 1e-10. The mean paired repetition change is -17.258%; a ratio of pooled means uses a different estimand. Lower aggregate cost does not establish that hold is preferable for every trajectory or thermal measure.

All 9,216 new runs complete; no thermal-limit violations are recorded. Missed deadlines total 79 for PI, 22 for PINO, 9,091 for each FV policy, and zero for Shared/Small. The inference-failure counter is initialized but not incremented in the harness, so no measured inference-failure rate is reported.

Executable compact checks

reproduce_monte_carlo.py checks the 9,456 saved costs and their saved components, full 64-by-six-by-24 roster, 384 group means, 18 distributions, 39 ranking events and 64 paired effects. This is saved-metric arithmetic, with no new inference or plant execution. The source map identifies every manuscript table and figure.

Full-archive audit receipts

The separate author audit reconstructs every cost from quadrature, applied commands and final fields (maximum error 6.67e-16), verifies archive CRC and file seals, and matches all 1,536 realized FV schedules. Deterministic resampling is checked for 384 selected cases across all six conditions and 64 repetitions. Full raw records and empirical pools are omitted from this compact release; those raw checks cannot be independently rerun from the saved summaries alone.